CS316: Classification (Compressed Notes)

Dr. Heyfa Ammar's lecture, slide order

1. What classification is

- Sorting observations into **categories** using their features. Y is **categorical**: one of c classes {0, ..., c-1}.
- **Framework:** Features X (age, BP, smoking) → Classifier g (fitted to data) → Class y-hat = g(x).
- **Supervised:** g is learned from data where the true class is known. (Clustering = unsupervised: no labels, groups similar observations.)
- **Binary** classification = 2 classes. Regression predicts a number; classification predicts a category.
- Points close to the decision boundary are the most **uncertain**.

2. Why accuracy is not enough (fingerprint scanners)

- Scanner A 99.55% vs B 99.54%: accuracy alone cannot decide. **Which kind of mistake** matters (letting an intruder in vs rejecting a soldier).

3. Expected loss (risk)

- **l(g) = E[Loss(Y, g(X))]**. Loss = price of one prediction; risk = average price over all cases; goal = the g with the smallest risk.

4. Zero-one loss

- **Loss(y, y-hat) = I{y ≠ y-hat}**: correct = 0, mistake = 1 (I = indicator: 1 if true, 0 if false).
- So the risk = the **probability of misclassifying** a new case. Example: 20 wrong out of 100 = 0.20.

5. Optimal (Bayes) classifier

- **$g^*(x) = \text{argmax}_y P[Y = y \mid X = x]$**: predict the class with the **highest probability** given the features.
- Steps: condition on x → score each class with P(Y = y | x) → choose the winner. Example: 0.2 / 0.7 / 0.1 → class 1.
- Optimal because always choosing the most likely class gives the lowest chance of a mistake. Ties are possible.
- Only **which class is largest** matters, not the whole f(y|x). The rule splits the feature space into **c regions**, one per class.

6. Bayes error rate

- **$l(g^*) = P[Y \neq g^*(X)]$**: the error of the BEST possible classifier.
- Where classes **overlap**, even g* errs (the shaded area). It is the **irreducible risk** (natural randomness), like Var(ε) in chapter 3. More training cannot remove it.
- The **decision boundary** is where the class probabilities cross.

7. Pre-classifier

- Outputs a real-valued **score** first. For labels -1/+1, score in [-1, 1], classifier = **sign(g)**: -0.4 → -1; 0.7 → +1.
- Score is continuous; the final output must be **discrete**. A score near 0 = near the boundary = uncertain.

8. Ways to fit a classifier

- Bayesian classification; linear and quadratic discriminant analysis; logistic classification; K-nearest neighbors; support vector machines; classification trees; neural networks. The lecture is about **judging** them.

9. Training and test sets

- **Fit** by minimizing the training loss $(1/n) \Sigma \text{Loss}(y_i, g(x_i))$. **Judge** on an independent test set: training loss is a poor (optimistic) estimate of the risk.

10. Confusion matrix and loss matrix

- **M(j, k)** = times actual class j was predicted as k. **Rows = actual, columns = predicted, diagonal = correct.**
- Zero-one loss matrix L: 0 on the diagonal, 1 elsewhere.
- **loss = $(1/n) \Sigma$ [L x M](j, k) = 1 - tr(M)/n** ([L x M] = multiply cell by cell, then add); tr(M) = sum of the diagonal = number correct.

11. Unequal costs (clinic: false alarm = 1, missed sick = 10, n = 1500)

Cell	Count	Cost	Contribution
TN healthy → healthy	1033	0	0
TP sick → sick	240	0	0
FP healthy → sick	91	1	91
FN sick → healthy	136	10	1360

- Weighted loss = (91 + 1360)/1500 = **0.967**; zero-one loss = (91 + 136)/1500 = **0.151**.
- Missed patients: 9% of the mistake count, 94% of the cost. Weighting tells the model which mistakes are dangerous.

12. Three-class example (Dog / Cat / Possum, n = 135)

Actual / Predicted	Dog	Cat	Possum
Dog	**30**	2	6
Cat	8	**22**	15
Possum	7	4	**41**

- Correct = 30 + 22 + 41 = 93; error = 1 - 93/135 = **31.1%**; accuracy = 68.9%.
- Biggest mistake: **Cat → Possum (15)**. Fix: add features that separate cats from possums.

13. The four counts for class j (one-vs-rest)

Count	Meaning	Where	Formula
TP	Actual j, predicted j	Diagonal cell	M(j, j)
FN	Actual j, predicted other	Rest of j's ROW	Σ M(j, k), k ≠ j
FP	Actual other, predicted j	Rest of j's COLUMN	Σ M(k, j), k ≠ j
TN	Neither actual j nor predicted j	Everything outside j's row and column	n - fn - fp - tp

- **Memory aid:** false positives live in the column; false negatives live in the row.
- **Dog:** TP 30, FN 8, FP 15, TN 82. **Cat:** TP 22, FP 6, FN 23, TN 84. **Possum:** TP 41, FP 21, FN 11, TN 62. Each sums to 135.

14. Error and accuracy

- **error(j) = (fp + fn)/n**; **accuracy(j) = (tp + tn)/n**; accuracy = 1 - error.
- With 2 classes, one class's FP are the other's FN, so the error is the same whichever class is 'positive'. Example: 20 sick→healthy and 30 healthy→sick give error 50/500 either way.

15. Exam traps

- High accuracy can be useless when one class dominates (always predicting 'healthy' on 95% healthy data = 95% accuracy, zero sick patients found).
- Same accuracy, different FN/FP: choose by the cost of the dangerous mistake (in medicine, FN is usually worse).
- A high P(class | x) means the classifier is confident about that case, not that the model is 'optimal'.
- Report the test error, not the training error.